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ORGANOMETALLIC COMPOUND, ORGANIC LIGHT-EMITTING DEVICE INCLUDING THE SAME, AND ELECTRONIC APPARATUS INCLUDING THE ORGANIC LIGHT-EMITTING DEVICE

Abstract

An organometallic compound represented by Formula 1:

M.sub.1(L.sub.1).sub.n1(L.sub.2).sub.n2 Formula 1 wherein M.sub.1 is a transition metal, L.sub.1 is a ligand represented by Formula 1A, L.sub.2 is a ligand represented by Formula 1B, and n1 and n2 are each independently 1 or 2:

##STR00001## wherein X.sub.2 is C(R.sub.2) or N, X.sub.3 is C(R.sub.3) or N, X.sub.4 is C(R.sub.4) or N, X.sub.5 is C(R.sub.5) or N, X.sub.6 is C(R.sub.6) or N, and X.sub.7 is C(R.sub.7) or N, wherein at least one of X.sub.2 to X.sub.7 is not N; at least one of R.sub.2 to R.sub.7 is —F, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3); or a C.sub.1-C.sub.60 alkyl, a C.sub.2-C.sub.60 alkenyl, a C.sub.2-C.sub.60 alkynyl, a C.sub.3-C.sub.10 cycloalkyl, a C.sub.1-C.sub.10 heterocycloalkyl, a C.sub.6-C.sub.60 aryl, or a C.sub.1-C.sub.60 heteroaryl, each substituted with at least one of —F, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —Ge(Q.sub.31)(Q.sub.32)(Q.sub.33), or a combination thereof, and the remaining substituents are as described herein.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and the benefit of Korean Patent Application No. 10-2024-0030111, filed on Feb. 29, 2024, in the Korean Intellectual Property Office, and all the benefits accruing therefrom under 35 U.S.C. § 119, the entire content of which is incorporated by reference herein.

BACKGROUND

1. Field

[0002] The disclosure relates to an organometallic compound, an organic light-emitting device including the same, and an electronic apparatus including the organic light-emitting device.

2. Description of the Related Art

[0003] Organic light-emitting devices (OLEDs) are self-emissive devices, have excellent characteristics in terms of viewing angles, response times, luminance, driving voltage, and response speed. OLEDs can also produce full-color images.

[0004] For example, an organic light-emitting device may include an anode, a cathode, and an organic layer arranged between the anode and the cathode and including an emission layer. A hole transport region may be arranged between the anode and the emission layer, and an electron transport region may be arranged between the emission layer and the cathode. Holes provided from the anode move toward the emission layer through the hole transport region, and electrons provided from the cathode move toward the emission layer through the electron transport region. The holes and the electrons recombine in the emission layer to produce excitons. The excitons transition from an excited state to a ground state, thereby generating light.

SUMMARY

[0005] Provided are an organometallic compound, an organic light-emitting device including the same, and an electronic apparatus including the organic light-emitting device.

[0006] Additional aspects will be set forth in part in the detailed description which follows and, in part, will be apparent from the detailed description, or may be learned by practice of the presented exemplary embodiments herein.

[0007] According to an aspect, provided is an organometallic compound represented by Formula 1.

M.sub.1(L.sub.1).sub.n1(L.sub.2).sub.n2 Formula 1

wherein, in Formula 1, [0008] M.sub.1 is a transition metal, [0009] L.sub.1 is a ligand represented by Formula 1A, [0010] L.sub.2 is a ligand represented by Formula 1B, and [0011] n1 and n2 are each independently 1 or 2,

##STR00002##

wherein, in Formulae 1A and 1B, [0012] X.sub.1 is C or N, [0013] X.sub.2 is C(R.sub.2) or N, X.sub.3 is C(R.sub.3) or N, X.sub.4 is C(R.sub.4) or N, X.sub.5 is C(R.sub.5) or N, X.sub.6 is C(R.sub.6) or N, and X.sub.7 is C(R.sub.7) or N, wherein at least one of X.sub.2 to X.sub.7 is not N, [0014] a1 is 1 or 2, [0015] R.sub.2 to R.sub.7, R.sub.20, and R.sub.31 to R.sub.33 are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF.sub.5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl

group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.1-C.sub.60 alkylthio group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.7-C.sub.60 alkyl aryl group, a substituted or unsubstituted C.sub.7-C.sub.60 aryl alkyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkyl heteroaryl group, a substituted or unsubstituted C.sub.2-C.sub.60 heteroaryl alkyl group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryloxy group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.4)(Q.sub.5), —B(Q.sub.6)(Q.sub.7), —P(Q.sub.8)(Q.sub.9), or —P(=O)(Q.sub.8)(Q.sub.9), [0016] at least one of R.sub.2 to R.sub.7 is: [0017] —F, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3), or [0018] a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.6-C.sub.60 aryl group, or a C.sub.1-C.sub.60 heteroaryl group, each substituted with at least one of —F, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —Ge(Q.sub.31)(Q.sub.32)(Q.sub.33), or a combination thereof, [0019] b20 is an integer from 1 to 5, [0020] two or more of R.sub.2 to R.sub.7 are optionally linked to each other to form a C.sub.5-C.sub.30 carbocyclic group that is unsubstituted or substituted with at least one R.sub.10a, or two or more of R.sub.2 to R.sub.7 are optionally linked to each other to form a C.sub.1-C.sub.30 heterocyclic group that is unsubstituted or substituted with at least one R.sub.10a, [0021] two or more of a plurality of R.sub.20 groups are optionally linked to each other to form a C.sub.5-C.sub.30 carbocyclic group that is unsubstituted or substituted with at least one R.sub.10a, or two or more of the plurality of R.sub.20 groups are optionally linked to each other to form a C.sub.1-C.sub.30 heterocyclic group that is unsubstituted or substituted with at least one R.sub.10a, [0022] R.sub.10a is as defined for R.sub.1, [0023] * and *' each indicate a binding site to M.sub.1, [0024] a substituent of the substituted C.sub.5-C.sub.30 carbocyclic group, the substituted C.sub.1-C.sub.30 heterocyclic group, the substituted C.sub.1-C.sub.60 alkyl group, the substituted C.sub.2-C.sub.60 alkenyl group, the substituted C.sub.2-C.sub.60 alkynyl group, the substituted C.sub.1-C.sub.60 alkoxy group, the substituted C.sub.1-C.sub.60 alkylthio group, the substituted C.sub.1-C.sub.60 alkylthio group, the substituted C.sub.3-C.sub.10 cycloalkyl group, the substituted C.sub.1-C.sub.10 heterocycloalkyl group, the substituted C.sub.3-C.sub.10 cycloalkenyl group, the substituted C.sub.1-C.sub.10 heterocycloalkenyl group, the substituted C.sub.6-C.sub.60 aryl group, the substituted C.sub.7-C.sub.60 alkyl aryl group, the substituted C.sub.7-C.sub.60 aryl alkyl group, the substituted C.sub.6-C.sub.60 aryloxy group, the substituted C.sub.6-C.sub.60 arylthio group, the substituted C.sub.1-C.sub.60 heteroaryl group, the substituted C.sub.2-C.sub.60 alkyl heteroaryl group, the substituted C.sub.2-C.sub.60 heteroaryl alkyl group, the substituted C.sub.1-C.sub.60 heteroaryloxy group, the substituted C.sub.1-C.sub.60 heteroarylthio group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group is each independently: [0025] deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, or a C.sub.1-C.sub.60 alkylthio group, [0026] a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, or a C.sub.1-C.sub.60 alkylthio group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a

hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.7-C.sub.60 alkyl aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.2-C.sub.60 alkyl heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —Ge(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.14)(Q.sub.15), [0027] —B(Q.sub.16)(Q.sub.17), —P(Q.sub.18)(Q.sub.19), —P(=O)(Q.sub.18)(Q.sub.19), or a combination thereof, [0028] a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.7-C.sub.60 alkyl aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.2-C.sub.60 alkyl heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group, [0029] a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.7-C.sub.60 alkyl aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.2-C.sub.60 alkyl heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.1-C.sub.60 alkylthio group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.7-C.sub.60 alkyl aryl group, a C.sub.7-C.sub.60 aryl alkyl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.2-C.sub.60 alkyl heteroaryl group, a C.sub.2-C.sub.60 heteroaryl alkyl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —Ge(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.24)(Q.sub.25), —B(Q.sub.26)(Q.sub.27), [0030] —P(Q.sub.28)(Q.sub.29), —P(=O)(Q.sub.28)(Q.sub.29), or a combination thereof, or [0031] —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —Ge(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.34)(Q.sub.35), —B(Q.sub.36)(Q.sub.37), —P(Q.sub.38)(Q.sub.39), or [0032] —P(=O)(Q.sub.38)(Q.sub.39), and [0033] Q.sub.1 to Q.sub.9, Q.sub.11 to Q.sub.19, Q.sub.21 to Q.sub.29, and Q.sub.31 to Q.sub.39 are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF.sub.5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.1-C.sub.60 alkylthio group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.7-C.sub.60 alkyl aryl group, a substituted or unsubstituted C.sub.7-C.sub.60 aryl alkyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or

unsubstituted C.sub.1-C.sub.10 heteroaryl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkyl heteroaryl group, a substituted or unsubstituted C.sub.2-C.sub.60 heteroaryl alkyl group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryloxy group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, or a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

[0034] According to another aspect, an organic light-emitting device includes a first electrode, a second electrode, and an organic layer arranged between the first electrode and the second electrode, wherein the organic layer includes an emission layer, and wherein the organic layer further includes at least one of the organometallic compounds represented by Formula 1.

[0035] The organometallic compound represented by Formula 1 may be included in the emission layer in the organic layer, and the organometallic compound included in the emission layer may act as a dopant.

Description

BRIEF DESCRIPTION OF THE DRAWING

[0036] The above and other aspects, features, and advantages of certain exemplary embodiments of the disclosure will be more apparent from the following detailed description taken in conjunction with the accompanying drawing, in which the FIGURE is a schematic cross-sectional view of an organic light-emitting device according to one or more embodiments.

DETAILED DESCRIPTION

[0037] Reference will now be made in detail to exemplary embodiments, examples of which are illustrated in the accompanying drawing, wherein like reference numerals refer to like elements throughout the specification. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

[0038] The terminology used herein is for the purpose of describing one or more exemplary embodiments only and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The term “or” means “and/or.” It will be further understood that the terms “comprises” and/or “comprising,” or “includes” and/or “including” when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof.

[0039] It will be understood that, although the terms first, second, third etc. may be used herein to describe various elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer, or section from another element, component, region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of the present embodiments.

[0040] Exemplary embodiments are described herein with reference to cross section illustrations that are schematic illustrations of idealized embodiments. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments described herein should not be construed as limited to the particular shapes of regions as illustrated herein but are to include deviations in shapes that result, for example, from manufacturing. For example, a region illustrated or described as flat may, typically, have rough and/or nonlinear features. Moreover, sharp angles that are illustrated may be rounded. Thus, the regions illustrated in the figures are schematic in nature and their shapes are not intended to illustrate the precise

shape of a region and are not intended to limit the scope of the present claims.

[0041] It will be understood that when an element is referred to as being “on” another element, it can be directly in contact with the other element or intervening elements may be present therebetween. In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present.

[0042] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this general inventive concept belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0043] “About” or “approximately” as used herein is inclusive of the stated value and means within an acceptable range of deviation for the particular value as determined by one of ordinary skill in the art, considering the measurement in question and the error associated with measurement of the particular quantity (i.e., the limitations of the measurement system). For example, “about” can mean within one or more standard deviations, or within $\pm 30\%$, 20% , 10% , 5% of the stated value.

[0044] An organometallic compound according to an aspect is represented by Formula 1:

M.sub.1(L.sub.1).sub.n1(L.sub.2).sub.n2 Formula 1

wherein, in Formula 1, M.sub.1 is a transition metal.

[0045] For example, M.sub.1 may be a first-row transition metal of the Periodic Table of Elements, a second-row transition metal of the Periodic Table of Elements, or a third-row transition metal of the Periodic Table of Elements.

[0046] In one or more embodiments, M.sub.1 may be iridium (Ir), platinum (Pt), osmium (Os), palladium (Pd), gold (Au), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), thulium (Tm), or rhodium (Rh).

[0047] In one or more embodiments, M.sub.1 may be iridium (Ir), platinum (Pt), osmium (Os), or rhodium (Rh).

[0048] In one or more embodiments, M.sub.1 may be iridium (Ir).

[0049] In Formula 1, n1 is 1 or 2, and n2 is 1 or 2.

[0050] In one or more embodiments, a sum of n1 and n2 may be 2 or 3.

[0051] In one or more embodiments, M.sub.1 may be iridium (Ir), and the sum of n1 and n2 may be 3.

[0052] In one or more embodiments, M.sub.1 may be platinum (Pt), and the sum of n1 and n2 may be 2.

[0053] In Formula 1, L.sub.1 is a ligand represented by Formula 1A.

##STR00003##

[0054] In Formula 1A, X.sub.1 is C or N. For example, X.sub.1 may be N.

[0055] In Formula 1A, X.sub.2 is C(R.sub.2) or N, X.sub.3 is C(R.sub.3) or N, X.sub.4 is C(R.sub.4) or N, X.sub.5 is C(R.sub.5) or N, X.sub.6 is C(R.sub.6) or N, and X.sub.7 is C(R.sub.7) or N.

[0056] At least one of X.sub.2 to X.sub.7 is not N.

[0057] In one or more embodiments, in Formula 1A, X.sub.2 may be C(R.sub.2), X.sub.3 may be C(R.sub.3), X.sub.4 may be C(R.sub.4), X.sub.5 may be C(R.sub.5), X.sub.6 may be C(R.sub.6), and X.sub.7 may be C(R.sub.7).

[0058] In Formula 1A, a1 is 1 or 2. For example, a1 may be 1.

[0059] In one or more embodiments, L.sub.1 may be a ligand represented by any one of Formulae 1A-1 to 1A-3.

##STR00004##

[0060] In Formulae 1A-1 to 1A-3, [0061] X.sub.1 to X.sub.7, a1, R.sub.20, and b20 are each as described herein, and [0062] * and *' each indicate a binding site to Mi.

[0063] In one or more embodiments, L.sub.1 may be a ligand represented by Formula 1A(1).

##STR00005##

[0064] In Formula 1A(1), [0065] R.sub.2 to R.sub.7, a1, R.sub.20, and b20 are each as described herein, and [0066] * and *' each indicate a binding site to M.sub.1.

[0067] In one or more embodiments, L.sub.1 may be a ligand represented by one of Formulae 1A(1)-1 to 1A(1)-3.

##STR00006##

[0068] In Formulae 1A(1)-1 to 1A(1)-3, [0069] X.sub.1, R.sub.2 to R.sub.7, a1, R.sub.20, and b20 are each as described herein, and [0070] * and *' each indicate a binding site to M.sub.1.

[0071] In one or more embodiments, L.sub.1 may be a ligand represented by Formula 1A(2).

##STR00007##

[0072] In Formula 1A(2), [0073] X.sub.1 to X.sub.7, R.sub.20, and b20 are each as described herein, and [0074] * and *' each indicate a binding site to M.sub.1.

[0075] In one or more embodiments, L.sub.1 may be a ligand represented by one of Formulae 1A(2)-1 to 1A(2)-6.

##STR00008## ##STR00009##

[0076] In Formulae 1A(2)-1 to 1A(2)-6, [0077] X.sub.1, R.sub.2 to R.sub.7, a1, R.sub.20, and b20 are each as described in claim 1, and [0078] * and *' each indicate a binding site to M.sub.1. [0079] L.sub.2 in Formula 1 is a ligand represented by Formula 1B.

##STR00010##

[0080] In Formulae 1A and 1B, R.sub.2 to R.sub.7, R.sub.20, and R.sub.31 to R.sub.33 are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF.sub.5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, —SF.sub.5, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.1-C.sub.60 alkylthio group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.7-C.sub.60 alkyl aryl group, a substituted or unsubstituted C.sub.7-C.sub.60 aryl alkyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a C.sub.2-C.sub.60 alkyl heteroaryl group, a C.sub.2-C.sub.60 heteroaryl alkyl group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryloxy group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.4)(Q.sub.5), —B(Q.sub.6)(Q.sub.7), —P(Q.sub.8)(Q.sub.9), or —P(=O)(Q.sub.8)(Q.sub.9).

[0081] In Formula 1A, two or more of R.sub.2 to R.sub.7 are optionally linked to each other to form a C.sub.5-C.sub.30 carbocyclic group that is unsubstituted or substituted with at least one R.sub.10a, or two or more of R.sub.2 to R.sub.7 are optionally linked to each other to form a C.sub.1-C.sub.30 heterocyclic group that is unsubstituted or substituted with at least one R.sub.10a, wherein R.sub.10a is as defined for R.sub.1.

[0082] In Formula 1A, two or more of a plurality of R.sub.20 are optionally linked to each other to form a C.sub.5-C.sub.30 carbocyclic group that is unsubstituted or substituted with at least one R.sub.10a, or two or more of the plurality of R.sub.20 are optionally linked to each other to form a C.sub.1-C.sub.30 heterocyclic group that is unsubstituted or substituted with at least one R.sub.10a, wherein R.sub.10a is as defined for R.sub.1.

[0083] In one or more embodiments, in Formulae 1A and 1B, R.sub.2 to R.sub.7, R.sub.20, and R.sub.31 to R.sub.33 may each independently be: [0084] hydrogen, deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, —SF.sub.5, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20

alkoxy group, or a C.sub.1-C.sub.20 alkylthio group; [0085] a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, or a C.sub.1-C.sub.20 alkylthio group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.10 alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or a combination thereof; [0086] a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzoimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group; [0087] a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzoimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a C.sub.1-C.sub.20 alkylthio group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzoimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a

dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, or a combination thereof; or [0088] —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.4)(Q.sub.5), —B(Q.sub.6)(Q.sub.7), —P(Q.sub.8)(Q.sub.9), or [0089] —P(=O)(Q.sub.8)(Q.sub.9), and [0090] Q.sub.1 to Q.sub.9 may each independently be: [0091] —CH.sub.3, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CH.sub.2CH.sub.3, —CH.sub.2CD.sub.3, —CH.sub.2CD.sub.2H, —CH.sub.2CDH.sub.2, —CHDCH.sub.3, —CHDCD.sub.2H, —CHDCDH.sub.2, —CHDCD.sub.3, —CD.sub.2CD.sub.3, —CD.sub.2CD.sub.2H, or —CD.sub.2CDH.sub.2; [0092] an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group; or [0093] an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group, each substituted with at least one of deuterium, a C.sub.1-C.sub.10 alkyl group, a phenyl group, or a combination thereof.

[0094] In one or more embodiments, in Formulae 1A and 1B, R.sub.2 to R.sub.7, R.sub.20, and R.sub.31 to R.sub.33 may each independently be: [0095] hydrogen, deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, or a C.sub.1-C.sub.60 alkylthio group; [0096] a group represented by one of Formulae 9-1 to 9-39, 9-44 to 9-61, 9-201 to 9-237, 10-1 to 10-129, or 10-201 to 10-353; or [0097] —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3), or —N(Q.sub.4)(Q.sub.5):
##STR00011## ##STR00012## ##STR00013## ##STR00014## ##STR00015## ##STR00016##
##STR00017## ##STR00018## ##STR00019## ##STR00020## ##STR00021## ##STR00022##
##STR00023## ##STR00024## ##STR00025## ##STR00026## ##STR00027## ##STR00028##
##STR00029## ##STR00030## ##STR00031## ##STR00032## ##STR00033## ##STR00034##
##STR00035##
##STR00036## ##STR00037## ##STR00038## ##STR00039## ##STR00040## ##STR00041##
##STR00042## ##STR00043## ##STR00044## ##STR00045## ##STR00046## ##STR00047##
##STR00048## ##STR00049## ##STR00050## ##STR00051## ##STR00052## ##STR00053##

[0098] In Formulae 9-1 to 9-39, 9-44 to 9-61, 9-201 to 9-240, 10-1 to 10-129, and 10-201 to 10-350, * indicates a binding site to a neighboring atom, “Ph” indicates a phenyl group, “TMS” indicates a trimethylsilyl group, and “TMG” indicates a trimethylgermyl group.

[0099] In Formula 1A, at least one of R.sub.2 to R.sub.7 is: [0100] —F, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3); or [0101] a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.6-C.sub.6 aryl group, or a C.sub.1-C.sub.60 heteroaryl group, each substituted with at least one of —F, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —Ge(Q.sub.31)(Q.sub.32)(Q.sub.33), or a combination thereof.

[0102] In one or more embodiments, at least one of R.sub.2 to R.sub.7 may be: [0103] —F, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3); or [0104] a C.sub.1-C.sub.20 alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzoimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a

benzobazoyl group, a dibenzocarbazoyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group, each substituted with at least one of —F, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —Ge(Q.sub.31)(Q.sub.32)(Q.sub.33), or a combination thereof, and [0105] Q.sub.1 to Q.sub.3 and Q.sub.31 to Q.sub.33 may each independently be: [0106] —CH.sub.3, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CH.sub.2CH.sub.3, —CH.sub.2CD.sub.3, —CH.sub.2CD.sub.2H, —CH.sub.2CDH.sub.2, —CHDCH.sub.3, —CHDCD.sub.2H, —CHDCDH.sub.2, —CHDCD.sub.3, —CD.sub.2CD.sub.3, —CD.sub.2CD.sub.2H, or —CD.sub.2CDH.sub.2; [0107] an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group; or [0108] an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group, each substituted with at least one of deuterium, a C.sub.1-C.sub.10 alkyl group, a phenyl group, or a combination thereof.

[0109] In one or more embodiments, at least one of R.sub.2 to R.sub.7 may be: [0110] —F, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3);

[0111] a group represented by any one of Formulae 9-1 to 9-39 in which at least one hydrogen atom is substituted with —F, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3); or [0112] a group represented by one of Formulae 10-1 to 10-129 or 10-201 to 10-350 in which at least one hydrogen atom is substituted with —F, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3).

[0113] In one or more embodiments, at least one of R.sub.2 to R.sub.7 may be: [0114] —F, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3);

[0115] a C.sub.1-C.sub.20 alkyl group that is substituted with at least one —F; or [0116] a group represented by one of Formulae 2A-1 to 2A-3:

##STR00054##

[0117] In Formulae 2A-1 to 2A-3, [0118] R.sub.10 may be hydrogen, deuterium, —F, —Cl, —Br, —I, —SF.sub.5, a hydroxyl group, a cyano group, a nitro group, an amino group, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a C.sub.1-C.sub.10 alkyl group, a C.sub.2-C.sub.10 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.30 aryl group, a C.sub.1-C.sub.30 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, [0119] —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.4)(Q.sub.5), —B(Q.sub.6)(Q.sub.7), —P(Q.sub.8)(Q.sub.9), or —P(=O)(Q.sub.8)(Q.sub.9), [0120] b₁₀ may be an integer from 1 to 4, and [0121] Q.sub.1 to Q.sub.9 and Q.sub.31 to Q.sub.33 may each independently be: [0122] —CH.sub.3, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CH.sub.2CH.sub.3, —CH.sub.2CD.sub.3, —CH.sub.2CD.sub.2H, —CH.sub.2CDH.sub.2, —CHDCH.sub.3, —CHDCD.sub.2H, —CHDCDH.sub.2, —CHDCD.sub.3, —CD.sub.2CD.sub.3, —CD.sub.2CD.sub.2H, or —CD.sub.2CDH.sub.2; [0123] an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group; or [0124] an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group, each substituted with at least one of deuterium, a C.sub.1-C.sub.1a alkyl group, a phenyl group, or a combination thereof.

[0125] In one or more embodiments, R.sub.31 to R.sub.33 may each independently be a C.sub.1-C.sub.20 alkyl group, a C.sub.3-C.sub.10 cycloalkyl group, or a C.sub.6-C.sub.30 aryl group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.1a alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group,

an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or a combination thereof.

[0126] In one or more embodiments, R.sub.33 may be hydrogen or deuterium.

[0127] In Formula 1A, b20 is an integer from 1 to 5.

[0128] In Formulae 1A and 1B, * and *' each indicate a binding site to M.sub.1.

[0129] In one or more embodiments, the organometallic compound may be represented by one of Formulae 5-1 to 5-3.

##STR00055##

[0130] In Formulae 5-1 to 5-3, [0131] M.sub.1, n1, n2, X.sub.1, R.sub.2 to R.sub.7, a1, R.sub.20, b20, and R.sub.31 to R.sub.33 are each as described herein.

[0132] In one or more embodiments, in Formulae 5-1 to 5-3, at least one of R.sub.2 to R.sub.7 may be:

[0133] —F, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —

Ge(Q.sub.1)(Q.sub.2)(Q.sub.3); [0134] a C.sub.1-C.sub.20 alkyl group that is substituted with at least one —F; or [0135] a group represented by one of Formulae 2A-1 to 2A-3.

[0136] In one or more embodiments, [0137] R.sub.5 may be —F, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3); [0138] a1 may be 1;

[0139] R.sub.31 and R.sub.33 may each independently be a C.sub.1-C.sub.20 alkyl group, a C.sub.3-C.sub.10 cycloalkyl group, or a C.sub.6-C.sub.60 aryl group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.10 alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or a combination thereof; and [0140] R.sub.32 may be hydrogen or deuterium.

[0141] In one or more embodiments, the organometallic compound represented by Formula 1 may be at least one of Compounds 1 to 72:

##STR00056## ##STR00057## ##STR00058## ##STR00059## ##STR00060## ##STR00061##
##STR00062## ##STR00063## ##STR00064## ##STR00065## ##STR00066## ##STR00067##
##STR00068## ##STR00069## ##STR00070## ##STR00071## ##STR00072## ##STR00073##
##STR00074## ##STR00075##

[0142] In one or more embodiments, the organometallic compound represented by Formula 1 may be electrically neutral.

[0143] The organometallic compound represented by Formula 1 includes ligands having structures represented by Formula 1A and Formula 1B, wherein, in Formula 1A, a ligand including X.sub.1 includes —F, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3) as a substituent, and a naphthyl ligand includes at least one cyano group as a substituent. Due to this structure, the organometallic compound represented by Formula 1 has excellent luminescence characteristics, and thus may have characteristics suitable for use as a luminescent material with high color purity by controlling the emission wavelength range and full width at half maximum (FWHM).

[0144] Therefore, an electronic device, e.g., an organic light-emitting device, including at least one of the organometallic compounds represented by Formula 1 may have a low driving voltage, high maximum external quantum efficiency, and/or a low roll-off ratio.

[0145] In one or more embodiments, the maximum emission wavelength (emission peak wavelength, $\lambda_{\text{sub.max}}$) of the emission peak in an emission spectrum or an electroluminescence (EL) spectrum of the organometallic compound represented by Formula 1 may be about 590 nanometers (nm) to about 650 nm, about 600 nm to about 640 nm, or about 610 nm to about 630 nm.

[0146] Synthesis methods of the organometallic compound represented by Formula 1 may be recognizable by one of ordinary skill in the art and by referring to Synthesis Examples provided below.

[0147] The organometallic compound represented by Formula 1 is suitable for use in an organic layer of

an organic light-emitting device, for example, for use as a dopant in an emission layer of the organic layer. Thus, another aspect provides an organic light-emitting device that includes a first electrode; a second electrode; an organic layer arranged between the first electrode and the second electrode, wherein the organic layer includes an emission layer; and wherein the organic layer further includes at least one of the organometallic compounds represented by Formula 1.

[0148] The organic light-emitting device includes the organic layer including at least one of the organometallic compounds represented by Formula 1, and thus, the organic light-emitting device may have excellent driving voltage, excellent maximum external quantum efficiency, and/or excellent lifespan characteristics.

[0149] The organometallic compound represented by Formula 1 may be used between a pair of electrodes of the organic light-emitting device. For example, at least one of the organometallic compounds represented by Formula 1 may be included in the emission layer.

[0150] At this time, the organometallic compound(s) represented by Formula 1 may act as a dopant, and the emission layer may further include a host. When the emission layer further includes a host, the amount of the host in the emission layer may be greater than the amount of the at least one organometallic compound represented by Formula 1 in the emission layer, on a weight basis.

[0151] In one or more embodiments, the emission layer may emit a red light. For example, the emission layer may emit an orange light or red light having a maximum emission wavelength of about 590 nm to about 650 nm, about 600 nm to about 640 nm, or about 610 nm to about 630 nm.

[0152] The expression “(organic layer) includes at least one of the organometallic compound represented by Formula 1” as used herein may mean that the (organic layer) may include one kind of organometallic compound represented by Formula 1 or two or more different kinds of organometallic compounds, each represented by Formula 1.

[0153] For example, the organic layer may include, as the at least one organometallic compound represented by Formula 1, only Compound 1. In this embodiment, Compound 1 may be included in the emission layer of the organic light-emitting device. In one or more embodiments, the organic layer may include, as the at least one organometallic compound represented by Formula 1, Compound 1 and Compound 2. In this embodiment, Compound 1 and Compound 2 may exist in an identical layer (for example, Compound 1 and Compound 2 all may exist in an emission layer).

[0154] The first electrode may be an anode, which is a hole injection electrode, and the second electrode may be a cathode, which is an electron injection electrode; or the first electrode may be a cathode, which is an electron injection electrode, and the second electrode may be an anode, which is a hole injection electrode.

[0155] In one or more embodiments, in the organic light-emitting device, the first electrode may be an anode, and the second electrode may be a cathode, the organic layer may further include a hole transport region located between the first electrode and the emission layer, and an electron transport region located between the emission layer and the second electrode, the hole transport region may include a hole injection layer, a hole transport layer, an electron blocking layer, a buffer layer, or a combination thereof, and the electron transport region may include a hole blocking layer, an electron transport layer, an electron injection layer, or a combination thereof.

[0156] The term “organic layer” as used herein refers to a single layer and/or a plurality of layers that are located between the first electrode and the second electrode of the organic light-emitting device. The “organic layer” may include, in addition to an organic compound, an organometallic complex including metal.

[0157] The FIGURE is a schematic cross-sectional view of an organic light-emitting device **10** according to one or more embodiments. Hereinafter, the structure of an organic light-emitting device according to one or more embodiments and a method of manufacturing an organic light-emitting device according to one or more embodiments will be described in connection with FIGURE. The organic light-emitting device **10** includes a first electrode **11**, an organic layer **15**, and a second electrode **19**, which are sequentially stacked in the stated order.

[0158] A substrate may be additionally disposed under the first electrode **11** or on the second electrode **19**. For use as the substrate, any substrate that is used in organic light-emitting devices available in the

art may be used, and the substrate may be a glass substrate or a transparent plastic substrate, each having excellent mechanical strength, thermal stability, transparency, surface smoothness, ease of handling, and/or water resistance.

[0159] The first electrode **11** may be formed by, for example, depositing or sputtering, onto the substrate, a material for forming the first electrode **11**. The first electrode **11** may be an anode. The material for forming the first electrode **11** may be selected from materials with a high work function for easy hole injection. The first electrode **11** may be a reflective electrode, a semi-transmissive electrode, or a transmissive electrode. The material for forming the first electrode may be indium tin oxide (ITO), indium zinc oxide (IZO), tin oxide (SnO₂), or zinc oxide (ZnO). In one or more embodiments, the material for forming the first electrode may be metal, such as magnesium (Mg), aluminum (Al), silver (Ag), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), or magnesium-silver (Mg—Ag).

[0160] The first electrode **11** may have a single-layer structure or a multilayer structure including two or more layers. For example, the first electrode **11** may have a three-layered structure of ITO/Ag/ITO, but the structure of the first electrode **11** is not limited thereto.

[0161] The organic layer **15** is located on the first electrode **11**.

[0162] The organic layer **15** may include a hole transport region, an emission layer, and an electron transport region.

[0163] The hole transport region may be arranged between the first electrode **11** and the emission layer.

[0164] The hole transport region may include a hole injection layer, a hole transport layer, an electron blocking layer, a buffer layer, or a combination thereof.

[0165] The hole transport region may include only either a hole injection layer or a hole transport layer. In one or more embodiments, the hole transport region may have a hole injection layer/hole transport layer structure, or a hole injection layer/hole transport layer/electron blocking layer structure, wherein, for each structure, respective layers are sequentially stacked in this stated order from the first electrode **11**.

[0166] When the hole transport region includes a hole injection layer, the hole injection layer (HIL) may be formed on the first electrode **11** by using one or more suitable methods, for example, vacuum deposition, spin coating, casting, and/or Langmuir-Blodgett (LB) deposition, but embodiments are not limited thereto.

[0167] When a hole injection layer is formed by vacuum deposition, the deposition conditions may vary according to a material that is used to form the hole injection layer, and the structure and thermal characteristics of the hole injection layer. For example, the deposition conditions may include a deposition temperature of about 100° C. to about 500° C., a vacuum pressure of about 10^{sup}−8 torr to about 10^{sup}−3 torr, and a deposition rate of about 0.01 angstroms per second (Å/sec) to about 100 Å/sec. However, the deposition conditions are not limited thereto.

[0168] When the hole injection layer is formed using spin coating, coating conditions may vary according to the material used to form the hole injection layer, and the structure and thermal properties of the hole injection layer. For example, a coating speed may be from about 2,000 revolutions per minute (rpm) to about 5,000 rpm, and a temperature at which a heat treatment is performed to remove a solvent after coating may be from about 80° C. to about 200° C. However, the coating conditions are not limited thereto.

[0169] The conditions for forming a hole transport layer and an electron blocking layer may be understood by referring to conditions for forming the hole injection layer.

[0170] The hole transport region may include, for example, at least one of 4,4',4''-tris(3-methylphenylphenylamino)triphenylamine (m-MTDATA), 4,4',4''-tris(N,N-diphenylamino)triphenylamine (TDATA), 4,4',4''-tris{N-(2-naphthyl)-N-phenylamino}-triphenylamine (2-TNATA), N,N'-di(1-naphthyl)-N,N'-diphenylbenzidine (NPB), β-NPB, N,N'-bis(3-methylphenyl)-N,N'-diphenyl-[1,1-biphenyl]-4,4'-diamine (TPD), Spiro-TPD, Spiro-NPB, methylated NPB, 4,4'-cyclohexylidene bis[N,N-bis(4-methylphenyl)benzenamine](TAPC), 4,4'-bis[N,N'-(3-tolyl)amino]-3,3'-dimethylbiphenyl (HMTDPD), 4,4',4''-tris(N-carbazolyl)triphenylamine (TCTA), polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA), poly(3,4-ethylenedioxythiophene)/poly(4-

styrenesulfonate) (PEDOT/PSS), polyaniline/camphor sulfonic acid (PANI/CSA), polyaniline/poly(4-styrenesulfonate) (PANI/PSS), a compound represented by Formula 201, or a compound represented by Formula 202, but embodiments are not limited thereto:

##STR00076## ##STR00077## ##STR00078## ##STR00079##

wherein, in Formula 201, Ar.sub.101 and Ar.sub.102 may each independently be: [0171] a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylene group, a heptalenylene group, an acenaphthylene group, a fluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylenylene group, a naphthacenylene group, a picenylene group, a perylenylene group, or a pentacenylene group; or [0172] a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylene group, a heptalenylene group, an acenaphthylene group, a fluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylenylene group, a naphthacenylene group, a picenylene group, a perylenylene group, or a pentacenylene group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.1-C.sub.60 alkylthio group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.7-C.sub.60 alkyl aryl group, a C.sub.7-C.sub.60 aryl alkyl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.2-C.sub.60 alkyl heteroaryl group, a C.sub.2-C.sub.60 heteroaryl alkyl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, or a combination thereof. [0173] xa and xb in Formula 201 may each independently be an integer from 0 to 5, or 0, 1, or 2. For example, xa may be 1 and xb may be 0, but embodiments are not limited thereto. [0174] R.sub.101 to R.sub.108, R.sub.111 to R.sub.119, and R.sub.121 to R.sub.124 in Formulae 201 and 202 may each independently be: [0175] hydrogen, deuterium, —F, —Cl, —Br, —I, —SF.sub.5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.10 alkyl group (e.g., a methyl group, an ethyl group, a propyl group, a butyl group, pentyl group, a hexyl group, or the like), a C.sub.1-C.sub.10 alkoxy group (e.g., a methoxy group, an ethoxy group, a propoxy group, a butoxy group, a pentoxy group, or the like), or a C.sub.1-C.sub.10 alkylthio group; [0176] a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a C.sub.1-C.sub.10 alkylthio group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, or a combination thereof; [0177] a phenyl group, a naphthyl group, an anthracenyl group, a fluorenyl group, or a pyrenyl group; or [0178] a phenyl group, a naphthyl group, an anthracenyl group, a fluorenyl group, or a pyrenyl group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a C.sub.1-C.sub.10 alkylthio group, or a combination thereof, but embodiments are not limited thereto. [0179] R.sub.109 in Formula 201 may be: [0180] a phenyl group, a naphthyl group, an anthracenyl group, or a pyridinyl group; or [0181] a phenyl group, a naphthyl group, an anthracenyl group, or a pyridinyl group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or

a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a C.sub.1-C.sub.20 alkylthio group, a phenyl group, a naphthyl group, an anthracenyl group, a pyridinyl group, or a combination thereof.

[0182] In one or more embodiments, the compound represented by Formula 201 may be represented by Formula 201A, but embodiments are not limited thereto:

##STR00080##

wherein, in Formula 201A, R.sub.101, R.sub.111, R.sub.112, and R.sub.109 are each as described herein.

[0183] For example, the compound represented by Formula 201 and the compound represented by Formula 202 may include at least one of Compounds HT1 to HT20, but embodiments are not limited thereto:

##STR00081## ##STR00082## ##STR00083##

[0184] The thickness of the hole transport region may be about 100 Å to about 10,000 Å, for example, about 100 Å to about 1,000 Å. When the hole transport region includes at least one of a hole injection layer and a hole transport layer, the thickness of the hole injection layer may be about 50 Å to about 10,000 Å, for example, about 100 Å to about 1,000 Å, and the thickness of the hole transport layer may be about 50 Å to about 2,000 Å, for example, about 100 Å to about 1,500 Å. When the thicknesses of the hole transport region, the hole injection layer and the hole transport layer are within these ranges, satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.

[0185] The hole transport region may further include, in addition to these materials, a charge-generation material for the improvement of conductive properties. The charge-generation material may be homogeneously or non-homogeneously dispersed in the hole transport region.

[0186] The charge-generation material may be, for example, a p-dopant. The p-dopant may be one of a quinone derivative, a metal oxide, or a cyano group-containing compound, but embodiments are not limited thereto. For example, the p-dopant may include a quinone derivative, such as tetracyanoquinodimethane (TCNQ), 2,3,5,6-tetrafluoro-tetracyano-1,4-benzoquinodimethane (F4-TCNQ), 1,3,4,5,7,8-hexafluorotetracyanonaphthoquinodimethane (F6-TCNNQ), or the like; a metal oxide, such as a tungsten oxide, a molybdenum oxide, or the like; or a cyano group-containing compound, such as one or more of Compounds HT-D1 or F12, but embodiments are not limited thereto:

##STR00084##

[0187] The hole transport region may further include a buffer layer.

[0188] The buffer layer may compensate for an optical resonance distance according to a wavelength of light emitted from the emission layer, and thus, efficiency of a formed organic light-emitting device may be improved.

[0189] Then, an emission layer (EML) may be formed on the hole transport region by vacuum deposition, spin coating, casting, LB deposition, or the like, but embodiments are not limited thereto. When the emission layer is formed by vacuum deposition or spin coating, the deposition or coating conditions may be similar to those applied in forming the hole injection layer, though the deposition or coating conditions may vary according to a material that is used to form the emission layer.

[0190] Meanwhile, when the hole transport region includes an electron blocking layer, a material for forming the electron blocking layer may be selected from materials for the hole transport region described herein and host materials as explained herein. However, the material for forming the electron blocking layer is not limited thereto. For example, when the hole transport region includes an electron blocking layer, a material for the electron blocking layer may be mCP, as described herein.

[0191] The emission layer may include a host and a dopant, and the dopant may include at least one of the organometallic compounds represented by Formula 1.

[0192] The host may include at least one of 1,3,5-tri(1-phenyl-1H-benzo[d]imidazol-2-yl)benzene (TPBi), 3-tert-butyl-9,10-di(naphth-2-yl)anthracene (TBADN), 9,10-di(naphthalene-2-yl)anthracene (ADN) (also referred to as "DNA"), 4,4'-bis(N-carbazolyl)-1,1'-biphenyl (CBP), 4,4'-bis(9-carbazolyl)-2,2'-dimethyl-biphenyl (CDBP), 1,3,5-tris(carbazole-9-yl)benzene (TCP), 1,3-bis(N-carbazolyl)benzene (mCP), Compound H50, or Compound H51, but embodiments are not limited

thereto:

##STR00085##

[0193] In one or more embodiments, the host may include a compound represented by Formula 301, but embodiments are not limited thereto:

##STR00086##

wherein, in Formula 301, Ar.sub.111 and Ar.sub.112 may each independently be: [0194] a phenylene group, a naphthylene group, a phenanthrenylene group, or a pyrenylene group; or [0195] a phenylene group, a naphthylene group, a phenanthrenylene group, or a pyrenylene group, each substituted with at least one of a phenyl group, a naphthyl group, an anthracenyl group, or a combination thereof.

[0196] Ar.sub.113 to Ar.sub.116 in Formula 301 may each independently be: [0197] a C.sub.1-C.sub.10 alkyl group, a phenyl group, a naphthyl group, a phenanthrenyl group, or a pyrenyl group; or [0198] a phenyl group, a naphthyl group, a phenanthrenyl group, or a pyrenyl group, each substituted with at least one of a phenyl group, a naphthyl group, an anthracenyl group, or a combination thereof. [0199] g, h, i, and j in Formula 301 may each independently be an integer from 0 to 4, for example, 0, 1, or 2.

[0200] Ar.sub.113 to Ar.sub.116 in Formula 301 may each independently be: [0201] a C.sub.1-C.sub.10 alkyl group that is substituted with at least one of a phenyl group, a naphthyl group, an anthracenyl group, or a combination thereof; [0202] a phenyl group, a naphthyl group, an anthracenyl group, a pyrenyl group, a phenanthrenyl group, or a fluorenyl group; [0203] a phenyl group, a naphthyl group, an anthracenyl group, a pyrenyl group, a phenanthrenyl group, or a fluorenyl group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.1-C.sub.60 alkylthio group, a phenyl group, a naphthyl group, an anthracenyl group, a pyrenyl group, a phenanthrenyl group, a fluorenyl group, or a combination thereof; or [0204] a group of the structure:

##STR00087## [0205] but, embodiments are not limited thereto.

[0206] In one or more embodiments, the host may include a compound represented by Formula 302, but embodiments are not limited thereto:

##STR00088## [0207] Ar.sub.122 to Ar.sub.125 in Formula 302 are each as described herein in connection with Ar.sub.113 in Formula 301. [0208] Ar.sub.126 and Ar.sub.127 in Formula 302 may each independently be a C.sub.1-C.sub.10 alkyl group (for example, a methyl group, an ethyl group, a propyl group, or the like). [0209] k and i in Formula 302 may each independently be an integer from 0 to 4. For example, k and i may be 0, 1, or 2.

[0210] When the organic light-emitting device is a full-color organic light-emitting device, the emission layer may be patterned into a red emission layer, a green emission layer, and/or a blue emission layer. In one or more embodiments, due to a stacked structure including a red emission layer, a green emission layer, and/or a blue emission layer, the emission layer may emit white light, and various modifications may be made.

[0211] When the emission layer includes a host and a dopant, the amount of the dopant in the emission layer may be about 0.01 parts by weight to about 15 parts by weight, based on 100 parts by weight of the host, but embodiments are not limited thereto.

[0212] The thickness of the emission layer may be about 100 Å to about 1,000 Å, for example, about 200 Å to about 600 Å. When the thickness of the emission layer is within the range described above, excellent luminescence characteristics may be obtained without a substantial increase in driving voltage.

[0213] An electron transport region is disposed on the emission layer.

[0214] The electron transport region may include a hole blocking layer, an electron transport layer, an electron injection layer, or a combination thereof.

[0215] For example, the electron transport region may have a hole blocking layer/electron transport layer/electron injection layer structure, or an electron transport layer/electron injection layer structure, but the disclosure is not limited thereto. The electron transport layer may have a single-layered structure or a multilayer structure including two or more different materials.

[0216] Conditions for forming the hole blocking layer, the electron transport layer, and the electron injection layer which constitute the electron transport region may be understood by referring to the conditions for forming the hole injection layer.

[0217] When the electron transport region includes a hole blocking layer, the hole blocking layer may include, for example, at least one of 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), or bis(2-methyl-8-quinolinolato-N1,O8)-(1,1'-biphenyl-4-olato)aluminum (BALq), but embodiments are not limited thereto:

##STR00089##

[0218] The thickness of the hole blocking layer may be about 20 Å to about 1,000 Å, for example, about 30 Å to about 300 Å. When the thickness of the hole blocking layer is within the range as described above, excellent hole blocking characteristics may be obtained without a substantial increase in driving voltage.

[0219] The electron transport layer may further include at least one of 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), tris(8-hydroxy-quinolinato)aluminum (Alq.sub.3), bis(2-methyl-8-quinolinolato-N1,O8)-(1,1'-biphenyl-4-olato)aluminum (BALq), 3-(4-biphenyl)-4-phenyl-5-tert-butylphenyl-1,2,4-triazole (TAZ), or 4-(naphthalen-1-yl)-3,5-diphenyl-4H-1,2,4-triazole (NTAZ), but embodiments are not limited thereto:

##STR00090##

[0220] In one or more embodiments, the electron transport layer may include at least one of Compounds ET1 to ET25, but embodiments are not limited thereto:

##STR00091## ##STR00092## ##STR00093## ##STR00094## ##STR00095##

[0221] The thickness of the electron transport layer may be about 100 Å to about 1,000 Å, for example, about 150 Å to about 500 Å. When the thickness of the electron transport layer is within the range as described above, satisfactory electron transport characteristics may be obtained without a substantial increase in driving voltage.

[0222] Also, the electron transport layer may further include, in addition to the materials described above, a metal-containing material.

[0223] The metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 (lithium quinolate, LiQ) or ET-D2, but embodiments are not limited thereto:

##STR00096##

[0224] In addition, the electron transport region may include an electron injection layer that facilitates electron injection from the second electrode **19**.

[0225] The electron injection layer may include LiF, NaCl, CsF, Li.sub.2O, BaO, or a combination thereof.

[0226] The thickness of the electron injection layer may be about 1 Å to about 100 Å, for example, about 3 Å to about 90 Å. When the thickness of the electron injection layer is within the range as described above, satisfactory electron injection characteristics may be obtained without a substantial increase in driving voltage.

[0227] The second electrode **19** is located on the organic layer **15**. The second electrode **19** may be a cathode. A material for forming the second electrode **19** may be metal, an alloy, an electrically conductive compound, or a combination thereof, which have a relatively low work function. For example, lithium (Li), magnesium (Mg), aluminum (Al), silver (Ag), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), or magnesium-silver (Mg—Ag) may be used as the material for forming the second electrode **19**. In one or more embodiments, to manufacture a top-emission type light-emitting device, various modifications, such as formation of a transmissive second electrode **19** using ITO or IZO, is possible.

[0228] Hereinbefore, the organic light-emitting device has been described with reference to the FIGURE, but embodiments are not limited thereto.

[0229] According to another aspect, provided is a diagnostic composition including at least one of the organometallic compounds represented by Formula 1.

[0230] The organometallic compound represented by Formula 1 provides a high luminescence

efficiency. Accordingly, a diagnostic composition including at least one of the organometallic compounds represented by Formula 1 may have high diagnostic efficiency.

[0231] The diagnostic composition may be used in various applications including a diagnosis kit, a diagnosis reagent, a biosensor, or a biomarker, but embodiments are not limited thereto.

[0232] The term “C.sub.1-C.sub.60 alkyl group” as used herein refers to a linear or branched saturated aliphatic hydrocarbon monovalent group having 1 to 60 carbon atoms, and non-limiting examples thereof include a methyl group, an ethyl group, a propyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, a pentyl group, an isoamyl group, a hexyl group, or the like. The term “C.sub.1-C.sub.60 alkylene group” as used herein refers to a divalent group having the same structure as the C.sub.1-C.sub.60 alkyl group.

[0233] The term “C.sub.1-C.sub.60 alkoxy group” used herein refers to a monovalent group represented by —OA.sub.101 (wherein A.sub.101 is the C.sub.1-C.sub.60 alkyl group), and non-limiting examples thereof include a methoxy group, an ethoxy group, an isopropoxy group, or the like. The term “C.sub.1-C.sub.60 alkylthio group” as used herein refers to a monovalent group represented by —SA.sub.101 (wherein A.sub.101 is the C.sub.1-C.sub.60 alkyl group).

[0234] The term “C.sub.2-C.sub.60 alkenyl group” as used herein has a structure including at least one carbon-carbon double bond in the middle or at the terminus of the C.sub.2-C.sub.60 alkyl group, and non-limiting examples thereof include an ethenyl group, a propenyl group, a butenyl group, or the like. The term “C.sub.2-C.sub.60 alkenylene group” as used herein refers to a divalent group having the same structure as the C.sub.2-C.sub.60 alkenyl group.

[0235] The term “C.sub.2-C.sub.60 alkynyl group” as used herein has a structure including at least one carbon-carbon triple bond in the middle or at the terminus of the C.sub.2-C.sub.60 alkyl group, and non-limiting examples thereof include an ethynyl group, a propynyl group, or the like. The term “C.sub.2-C.sub.60 alkynylene group” as used herein refers to a divalent group having the same structure as the C.sub.2-C.sub.60 alkynyl group.

[0236] The term “C.sub.3-C.sub.10 cycloalkyl group” as used herein refers to a monovalent saturated hydrocarbon ring group having 3 to 10 carbon atoms, and non-limiting examples thereof include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, or the like. The term “C.sub.3-C.sub.10 cycloalkylene group” as used herein refers to a divalent group having the same structure as the C.sub.3-C.sub.10 cycloalkyl group.

[0237] The term “C.sub.1-C.sub.10 heterocycloalkyl group” as used herein refers to a monovalent ring group having at least one heteroatom selected from B, N, O, P, Si, S, Se, or Ge as a ring-forming atom and 1 to 10 carbon atoms as ring-forming atom(s), and non-limiting examples thereof include a tetrahydrofuranyl group, a tetrahydrothiophenyl group (tetrahydrothiophene group), or the like. The term “C.sub.1-C.sub.10 heterocycloalkylene group” as used herein refers to a divalent group having the same structure as the C.sub.1-C.sub.10 heterocycloalkyl group.

[0238] The term “C.sub.3-C.sub.10 cycloalkenyl group” as used herein refers to a monovalent ring group that includes 3 to 10 carbon atoms and at least one carbon-carbon double bond in the ring thereof and has no aromaticity, and non-limiting examples thereof include a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, or the like. The term “C.sub.3-C.sub.10 cycloalkenylene group” as used herein refers to a divalent group having the same structure as the C.sub.3-C.sub.10 cycloalkenyl group.

[0239] The term “C.sub.1-C.sub.10 heterocycloalkenyl group” as used herein refers to a monovalent ring group that has at least one heteroatom selected from B, N, O, P, Si, S, Se, and Ge as a ring-forming atom, 1 to 10 carbon atoms as ring-forming atom(s), and at least one double bond in the ring thereof. Non-limiting examples of the C.sub.1-C.sub.10 heterocycloalkenyl group include a 2,3-dihydrofuranyl group, a 2,3-dihydrothiophenyl group, or the like. The term “C.sub.1-C.sub.10 heterocycloalkenylene group” as used herein refers to a divalent group having the same structure as the C.sub.1-C.sub.10 heterocycloalkenyl group.

[0240] The term “C.sub.6-C.sub.60 aryl group” as used herein refers to a monovalent group having a carbocyclic aromatic ring system having 6 to 60 carbon atoms, and the term “C.sub.6-C.sub.60 arylene group” as used herein refers to a divalent group having a carbocyclic aromatic ring system having 6 to

60 carbon atoms. Non-limiting examples of the C.sub.6-C.sub.60 aryl group include a phenyl group, a naphthyl group, an anthracenyl group, a phenanthrenyl group, a pyrenyl group, a chrysenyl group, or the like. When the C.sub.6-C.sub.60 aryl group and the C.sub.6-C.sub.60 arylene group each include two or more rings, the two or more rings may be fused to each other.

[0241] The term “C.sub.7-C.sub.60 alkyl aryl group” as used herein refers to a C.sub.6-C.sub.60 aryl group that is substituted with at least one C.sub.1-C.sub.60 alkyl group. The term “C.sub.7-C.sub.60 aryl alkyl group” as used herein refers to a C.sub.1-C.sub.60 alkyl group that is substituted with at least one C.sub.6-C.sub.60 aryl group.

[0242] The term “C.sub.1-C.sub.60 heteroaryl group” as used herein refers to a monovalent group having a heteroaromatic ring system having at least one heteroatom selected from B, N, O, P, Si, S, Se, and Ge as a ring-forming atom and 1 to 60 carbon atoms as ring-forming atom(s). The term “C.sub.1-C.sub.60 heteroarylene group” as used herein refers to a divalent group having a heteroaromatic ring system having at least one heteroatom selected from B, N, O, P, Si, S, Se, and Ge as a ring-forming atom and 1 to 60 carbon atoms as ring-forming atom(s). Non-limiting examples of the C.sub.1-C.sub.60 heteroaryl group include a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, or the like. When the C.sub.1-C.sub.60 heteroaryl group and the C.sub.1-C.sub.60 heteroarylene group each include two or more rings, the two or more rings may be fused to each other.

[0243] The term “C.sub.2-C.sub.60 alkyl heteroaryl group” as used herein refers to a C.sub.1-C.sub.60 heteroaryl group that is substituted with at least one C.sub.1-C.sub.60 alkyl group. The term “C.sub.2-C.sub.60 heteroaryl alkyl group” as used herein refers to a C.sub.1-C.sub.60 alkyl group that is substituted with at least one C.sub.1-C.sub.60 heteroaryl group.

[0244] The term “C.sub.6-C.sub.60 aryloxy group” as used herein refers to —OA.sub.102 (wherein A.sub.102 is the C.sub.6-C.sub.60 aryl group), and the term “C.sub.6-C.sub.60 arylthio group” as used herein refers to —SA.sub.103 (wherein A.sub.103 is the C.sub.6-C.sub.60 aryl group).

[0245] The term “C.sub.1-C.sub.60 heteroaryloxy group” as used herein refers to —OA.sub.104 (wherein A.sub.104 is the C.sub.1-C.sub.60 heteroaryl group), and the term “C.sub.1-C.sub.60 heteroarylthio group” as used herein refers to —SA.sub.105 (wherein A.sub.105 is the C.sub.1-C.sub.60 heteroaryl group).

[0246] The term “monovalent non-aromatic condensed polycyclic group” as used herein refers to a monovalent group (for example, having 8 to 60 carbon atoms) having two or more rings condensed to each other, only carbon atoms as ring-forming atoms, and no aromaticity in its molecular structure when considered as a whole. Non-limiting examples of the monovalent non-aromatic condensed polycyclic group include a fluorenyl group or the like.

[0247] The term “divalent non-aromatic condensed polycyclic group” as used herein refers to a divalent group having the same structure as the monovalent non-aromatic condensed polycyclic group.

[0248] The term “monovalent non-aromatic condensed heteropolycyclic group” as used herein refers to a monovalent group having two or more rings condensed with each other, a heteroatom selected from B, N, O, P, Si, S, Se, and Ge, other than carbon atoms (for example, having 1 to 60 carbon atoms), as a ring-forming atom, and no aromaticity in its molecular structure when considered as a whole. Non-limiting examples of the monovalent non-aromatic condensed heteropolycyclic group includes a carbazolyl group or the like. The term “divalent non-aromatic condensed heteropolycyclic group” as used herein refers to a divalent group having the same structure as the monovalent non-aromatic condensed heteropolycyclic group.

[0249] The term “C.sub.5-C.sub.30 carbocyclic group” as used herein refers to a saturated or unsaturated ring group having, as a ring-forming atom, 5 to 30 carbon atoms only. The C.sub.5-C.sub.30 carbocyclic group may be a monocyclic group or a polycyclic group.

[0250] The term “C.sub.1-C.sub.30 heterocyclic group” as used herein refers to a saturated or unsaturated ring group having 1 to 30 carbon atoms as a ring-forming atom and at least one heteroatom selected from B, N, O, P, Si, S, Se, and Ge. The C.sub.1-C.sub.30 heterocyclic group may be a monocyclic group or a polycyclic group.

[0251] At least one substituent of the substituted C.sub.5-C.sub.30 carbocyclic group, the substituted

C.sub.1-C.sub.60 heterocycloalkyl group, the substituted C.sub.1-C.sub.60 alkyl group, the substituted C.sub.2-C.sub.60 alkenyl group, the substituted C.sub.2-C.sub.60 alkynyl group, the substituted C.sub.1-C.sub.60 alkoxy group, the substituted C.sub.1-C.sub.60 alkylthio group, the substituted C.sub.1-C.sub.60 alkylthio group, the substituted C.sub.3-C.sub.10 cycloalkyl group, the substituted C.sub.1-C.sub.10 heterocycloalkyl group, the substituted C.sub.3-C.sub.10 cycloalkenyl group, the substituted C.sub.1-C.sub.10 heterocycloalkenyl group, the substituted C.sub.6-C.sub.60 aryl group, the substituted C.sub.7-C.sub.60 alkyl aryl group, the substituted C.sub.7-C.sub.60 aryl alkyl group, the substituted C.sub.6-C.sub.60 aryloxy group, the substituted C.sub.6-C.sub.60 arylthio group, the substituted C.sub.1-C.sub.60 heteroaryl group, the substituted C.sub.2-C.sub.60 alkyl heteroaryl group, the substituted C.sub.2-C.sub.60 heteroaryl alkyl group, the substituted C.sub.1-C.sub.60 heteroaryloxy group, the substituted C.sub.1-C.sub.60 heteroarylthio group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group may be: [0252] deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, or a C.sub.1-C.sub.60 alkylthio group; [0253] a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, or a C.sub.1-C.sub.60 alkylthio group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.7-C.sub.60 alkyl aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.2-C.sub.60 alkyl heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —Ge(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.14)(Q.sub.15), [0254] —B(Q.sub.16)(Q.sub.17), —P(Q.sub.18)(Q.sub.19), —P(=O)(Q.sub.18)(Q.sub.19), or a combination thereof; [0255] a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.7-C.sub.60 alkyl aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.2-C.sub.60 alkyl heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group; [0256] a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.7-C.sub.60 alkyl aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.2-C.sub.60 alkyl heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1—Co alkoxy group, a C.sub.1-C.sub.60 alkylthio group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.7-C.sub.60 alkyl aryl group, a C.sub.7-C.sub.60 aryl alkyl

group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.2-C.sub.60 alkyl heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), [0257] —Ge(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.24)(Q.sub.25), —B(Q.sub.26)(Q.sub.27), —P(Q.sub.28)(Q.sub.29), —P(=O)(Q.sub.28)(Q.sub.29), or a combination thereof; or [0258] —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —Ge(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.34)(Q.sub.35), —B(Q.sub.36)(Q.sub.37), —P(Q.sub.38)(Q.sub.39), or [0259] —P(=O)(Q.sub.38)(Q.sub.39), and [0260] Q.sub.1 to Q.sub.9, Q.sub.11 to Q.sub.19, Q.sub.21 to Q.sub.29, and Q.sub.31 to Q.sub.39 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, —SF.sub.5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.1-C.sub.60 alkylthio group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.7-C.sub.60 alkyl aryl group, a substituted or unsubstituted C.sub.7-C.sub.60 aryl alkyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1—Co heteroaryl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkyl heteroaryl group, a substituted or unsubstituted C.sub.2-C.sub.60 heteroaryl alkyl group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryloxy group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, or a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

[0261] For example, Q.sub.1 to Q.sub.9, Q.sub.11 to Q.sub.19, Q.sub.21 to Q.sub.29, and Q.sub.31 to Q.sub.39 may each independently be a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.1-C.sub.60 alkylthio group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.7-C.sub.60 alkyl aryl group, a C.sub.7-C.sub.60 aryl alkyl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.2-C.sub.60 alkyl heteroaryl group, a C.sub.2-C.sub.60 heteroaryl alkyl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group, each unsubstituted or substituted with at least one of hydrogen, deuterium, —F, —Cl, —Br, —I, —SF.sub.5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof; deuterium, a C.sub.1-C.sub.60 alkyl group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.6-C.sub.60 aryl group, or a combination thereof.

[0262] Hereinafter, exemplary organometallic compounds represented by Formula 1 and organic light-emitting devices according to exemplary embodiments are described in further detail with reference to Synthesis Examples and Examples. However, embodiments are not limited to Synthesis Examples and Examples. The wording “B was used instead of A” used in describing Synthesis Examples means that an amount of A used was identical to an amount of B used, in terms of a molar equivalent.

EXAMPLES

Synthesis Example 1: Synthesis of Compound 1

##STR00097##

Synthesis of Intermediate L1

[0263] 2.0 grams (g) (8.48 millimoles (mmol)) 1-chloro-6-(trimethylsilyl)isoquinoline was mixed with

60 milliliters (mL) of tetrahydrofuran (THF) and 15 mL of water, and then 2.8 g (10.17 mmol) of 3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-naphthonitrile, 0.7 g (0.59 mmol) of (tetrakis(triphenylphosphine)palladium(0), Pd(PPh.sub.3).sub.4), and 2.9 g (21.1 mmol) of K.sub.2CO.sub.3 were added thereto. The resulting mixture was then heated under reflux for 24 hours. The resulting product obtained therefrom was subjected to an extraction process by using ethyl acetate and water to obtain an organic layer. The organic layer was then dried with magnesium sulfate, the solvent was removed under a reduced pressure, and the product was purified by liquid chromatography, to thereby obtain 2.2 g (yield of 75%) of Intermediate L1.

[0264] Liquid chromatography-mass spectrometry (LC-MS) m/z=353 (M+H).sup.+.

Synthesis of L1 dimer

[0265] 1.0 g (2.8 mmol) of Intermediate L1 and 0.47 g (1.3 mmol) of iridium chloride hydrate were mixed with 60 mL of ethoxyethanol and 15 mL of deionized (DI) water and then heated under reflux for 24 hours. The temperature of the resulting product obtained therefrom was lowered to room temperature, and the resulting solids were filtered and then sufficiently washed with water/methanol/hexane in the stated order. The solids obtained without separate filtration were dried in a vacuum oven, to thereby obtain 0.9 g of L1 dimer.

Synthesis of Compound 1

[0266] 0.9 g (0.48 mmol) of L1 dimer, 0.12 g (1.2 mmol) of acetylacetone, and 0.17 g (1.2 mmol) of K.sub.2CO.sub.3 were mixed with 30 mL of ethoxyethanol and then stirred at 110° C. for 18 hours. After the reaction was completed, the temperature was lowered to room temperature, and the resulting solids were filtered. The product was purified by liquid chromatography, to thereby obtain 0.5 g (yield of 50%) of Compound 1.

[0267] LC-MS m/z=995 (M+H).sup.+.

Synthesis Example 2: Synthesis of Compound 5

##STR00098##

Synthesis of Intermediate L5

[0268] 2.2 g (yield of 78%) of Intermediate L9 was synthesized in a similar manner as used to synthesize Intermediate L2 of Synthesis Example 1, except that 1-chloro-6-(trimethylgermyl)isoquinoline was used instead of 1-chloro-6-(trimethylsilyl)isoquinoline.

[0269] LC-MS m/z=399 (M+H).sup.+.

Synthesis of L5 Dimer

[0270] The L5 dimer was synthesized in a similar manner as used to synthesize L1 dimer of Synthesis Example 1, except that Intermediate L5 was used instead of Intermediate L1.

Synthesis of Compound 5

[0271] 5 g (yield of 48%) of Compound 5 was synthesized in a similar manner as used to synthesize Compound 1 of Synthesis Example 1, except that L5 dimer was used instead of L1 dimer.

[0272] LC-MS m/z=1087 (M+H).sup.+.

Synthesis Example 3: Synthesis of Compound 9

##STR00099##

Synthesis of Intermediate L9

[0273] 1.3 g (yield of 90%) of Intermediate L9 was synthesized in a similar manner as used to synthesize Intermediate L1 of Synthesis Example 1, except that 1-chloro-6-(trifluoromethyl)isoquinoline was used instead of 1-chloro-6-(trimethylsilyl)isoquinoline, and ethanol was used instead of THE as a solvent.

[0274] LC-MS m/z=349 (M+H).sup.+.

Synthesis of L9 Dimer

[0275] The L9 dimer was synthesized in a similar manner as used to synthesize L1 dimer of Synthesis Example 1, except that Intermediate L9 was used instead of Intermediate L1.

Synthesis of Compound 9

[0276] 0.3 g (yield of 35%) of Compound 9 was synthesized in a similar manner as used to synthesize Compound 1 of Synthesis Example 1, except that the L9 dimer was used instead of the L1 dimer.

[0277] LC-MS m/z=987 (M+H).sup.+.

Synthesis Example 4: Synthesis of Compound 28

##STR00100##

Synthesis of Intermediate L28

[0278] 0.9 g (yield of 75%) of Intermediate L28 was synthesized in a similar manner as used to synthesize Intermediate L1 of Synthesis Example 1, except that 1-chloro-6-(4-(trimethylgermyl)phenyl)isoquinoline was used instead of 4-chloro-7-methyl benzo[c][2,7]naphthyridine.

[0279] LC-MS m/z=474 (M+H).sup.+.

Synthesis of L28 Dimer

[0280] The L28 dimer was synthesized in a similar manner as used to synthesize L1 dimer of Synthesis Example 1, except that Intermediate L28 was used instead of Intermediate L1.

Synthesis of Compound 28

[0281] 0.2 g (yield of 30%) of Compound 28 was synthesized in a similar manner as used to synthesize Compound 1 of Synthesis Example 1, except that the L28 dimer was used instead of the L1 dimer.

[0282] LC-MS m/z=1237 (M+H).sup.+.

Example 1

[0283] As an anode, an ITO-patterned glass substrate was cut to a size of 50 millimeters (mm)×50 mm×0.5 mm, sonicated with isopropyl alcohol and DI water, each for 5 minutes, and then cleaned by irradiation of ultraviolet rays and exposure of ozone thereto for 30 minutes. The resultant glass substrate was loaded onto a vacuum deposition apparatus.

[0284] Compound HT3 and F12-P-Dopant were vacuum co-deposited on the anode at a weight ratio of 98:2 to form a hole injection layer having a thickness of 100 Å, and Compound HT3 was vacuum deposited on the hole injection layer to form a hole transport layer having a thickness of 1,600 Å.

[0285] Next, Compound RH3 (host) and Compound 1 (dopant) were co-deposited on the hole transport layer at a weight ratio of 97:3 to form an emission layer having a thickness of 400 Å.

[0286] Afterwards, Compound ETL and Liq-N-Dopant were co-deposited on the emission layer at a volume ratio of 50:50 to form an electron transport layer having a thickness of 350 Å, Liq-N-Dopant was vacuum deposited on the electron transport layer to form an electron injection layer having a thickness of 10 Å, and Al was vacuum deposited on the electron injection layer to form a cathode having a thickness of 1,000 Å, thereby completing manufacture of an organic light-emitting device.

##STR00101##

Examples 2 to 4 and Comparative Examples 1 to 4

[0287] Organic light-emitting devices were manufactured in a similar manner as in Example 1, except that, in forming the emission layer, each of compounds shown in Table 1 was used instead of Compound 1.

Evaluation Example 1: Evaluation of Characteristics of Organic Light-Emitting Devices

[0288] For the organic light-emitting devices of Examples 1 to 4 and Comparative Examples 1 to 4, driving voltage (volts, V), maximum value of external quantum efficiency (Max EQE, %), maximum emission wavelength (nm), full width at half maximum (FWHM, nm), and roll-off ratio (%) were evaluated, and the results thereof are shown in Table 1. A current voltmeter (Keithley 2400) and a luminance meter (Minolta Cs-1000A) were used as evaluation apparatuses. The roll-off ratio was calculated according to Equation 20.

[00001]

$$\text{Roll-off ratio} = \{1 - (\text{efficiency}(\text{at } 3, 500 \text{ cd/m}^2) / \text{maximum luminescence efficiency})\} \times 100\% \quad \text{Equation 20}$$

TABLE-US-00001	TABLE 1	Maximum Driving Voltage (V)	Max emission wavelength (nm)	Roll-off ratio (%)	Dopant	EQE (%)	emission wavelength (nm)	FWHM (nm)
Example 1	1	4.5	25	619	36	10	Example 2	5
Example 2	2	4.5	25	618	36	10	Example 3	9
Example 3	3	4.5	25	640	37	11	Example 4	28
Example 4	4	4.6	27	622	39	12	Comparative CE1	4.8
Comparative CE1	1	4.8	23	610	35	13	Example 1	4.9
Comparative CE2	2	4.9	23	650	36	14	Example 2	4.8
Comparative CE3	3	4.8	24	656	37	13	Example 3	4.8
Comparative CE4	4	4.8	20	651	38	15	Example 4	[00102]
[00102]							[00103]	
[00103]							[00104]	
[00104]							[00105]	
[00105]							[00106]	
[00106]							[00107]	
[00107]							[00108]	
[00108]							[00109]	
[00109]							[00110]	

[0289] From Table 1, it was confirmed that each of the organic light-emitting devices of Examples 1 to 4 had a low driving voltage, a high maximum external quantum efficiency, a narrow FWHM, and a low

roll-off ratio.

[0290] In addition, it was confirmed that the organic light-emitting devices of Examples 1 to 4 exhibited a lower driving voltage, an equal or greater maximum external quantum efficiency, and a lower roll-off ratio, compared to the organic light-emitting devices of Comparative Examples 1 to 4.

[0291] The organometallic compound represented by Formula 1 has excellent electrical characteristics, and thus, an electronic device, e.g., an organic light-emitting device, including at least one of the organometallic compounds represented by Formula 1 may have a low driving voltage, a high maximum external quantum efficiency, a narrow FWHM, and a low roll-off ratio. Therefore, a high-quality organic light-emitting device may be embodied by using the organometallic compound represented by Formula 1.

[0292] It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figure, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope as defined by the following claims.

Claims

1. An organometallic compound represented by Formula 1:

$M_{sub.1}(L_{sub.1})_{sub.n1}(L_{sub.2})_{sub.n2}$ Formula 1 wherein, in Formula 1, $M_{sub.1}$ is a transition metal, $L_{sub.1}$ is a ligand represented by Formula 1A, $L_{sub.2}$ is a ligand represented by Formula 1B, and $n1$ and $n2$ are each independently 1 or 2, ##STR00110## wherein, in Formulae 1A and 1B, $X_{sub.1}$ is C or N, $X_{sub.2}$ is $C(R_{sub.2})$ or N, $X_{sub.3}$ is $C(R_{sub.3})$ or N, $X_{sub.4}$ is $C(R_{sub.4})$ or N, $X_{sub.5}$ is $C(R_{sub.5})$ or N, $X_{sub.6}$ is $C(R_{sub.6})$ or N, and $X_{sub.7}$ is $C(R_{sub.7})$ or N, wherein at least one of $X_{sub.2}$ to $X_{sub.7}$ is not N, $a1$ is 1 or 2, $R_{sub.2}$ to $R_{sub.7}$, $R_{sub.20}$, and $R_{sub.31}$ to $R_{sub.33}$ are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF_{sub.5}, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C_{sub.1}-C_{sub.60} alkyl group, a substituted or unsubstituted C_{sub.2}-C_{sub.60} alkenyl group, a substituted or unsubstituted C_{sub.2}-C_{sub.60} alkynyl group, a substituted or unsubstituted C_{sub.1}-C_{sub.60} alkoxy group, a substituted or unsubstituted C_{sub.1}-C_{sub.60} alkylthio group, a substituted or unsubstituted C_{sub.3}-C_{sub.10} cycloalkyl group, a substituted or unsubstituted C_{sub.1}-C_{sub.10} heterocycloalkyl group, a substituted or unsubstituted C_{sub.3}-C_{sub.10} cycloalkenyl group, a substituted or unsubstituted C_{sub.1}-C_{sub.10} heterocycloalkenyl group, a substituted or unsubstituted C_{sub.6}-C_{sub.60} aryl group, a substituted or unsubstituted C_{sub.7}-C_{sub.60} alkyl aryl group, a substituted or unsubstituted C_{sub.7}-C_{sub.60} aryl alkyl group, a substituted or unsubstituted C_{sub.6}-C_{sub.60} aryloxy group, a substituted or unsubstituted C_{sub.6}-C_{sub.60} arylthio group, a substituted or unsubstituted C_{sub.1}-C_{sub.60} heteroaryl group, a substituted or unsubstituted C_{sub.2}-C_{sub.60} alkyl heteroaryl group, a substituted or unsubstituted C_{sub.2}-C_{sub.60} heteroaryl alkyl group, a substituted or unsubstituted C_{sub.1}-C_{sub.60} heteroaryloxy group, a substituted or unsubstituted C_{sub.1}-C_{sub.60} heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q_{sub.1})(Q_{sub.2})(Q_{sub.3}), —Ge(Q_{sub.1})(Q_{sub.2})(Q_{sub.3}), —N(Q_{sub.4})(Q_{sub.5}), —B(Q_{sub.6})(Q_{sub.7}), —P(Q_{sub.8})(Q_{sub.9}), or —P(=O)(Q_{sub.8})(Q_{sub.9}), at least one of $R_{sub.2}$ to $R_{sub.7}$ is: —F, —Si(Q_{sub.1})(Q_{sub.2})(Q_{sub.3}), or —Ge(Q_{sub.1})(Q_{sub.2})(Q_{sub.3}); or a C_{sub.1}-C_{sub.60} alkyl group, a C_{sub.2}-C_{sub.60} alkenyl group, a C_{sub.2}-C_{sub.60} alkynyl group, a C_{sub.3}-C_{sub.10} cycloalkyl group, a C_{sub.1}-C_{sub.10} heterocycloalkyl group, a C_{sub.6}-C_{sub.60} aryl group, or a C_{sub.1}-C_{sub.60} heteroaryl group, each substituted with at least one of —F, —Si(Q_{sub.31})(Q_{sub.32})(Q_{sub.33}), —Ge(Q_{sub.31})(Q_{sub.32})(Q_{sub.33}), or a combination thereof, $b20$ is an integer from 1 to 5, two or more of $R_{sub.2}$ to $R_{sub.7}$ are optionally linked to each other to form a C_{sub.5}-C_{sub.30} carbocyclic group

that is unsubstituted or substituted with at least one R.sub.10a, or two or more of R.sub.2 to R.sub.7 are optionally linked to each other to form a C.sub.1-C.sub.30 heterocyclic group that is unsubstituted or substituted with at least one R.sub.10a, two or more of a plurality of R.sub.20 are optionally linked to each other to form a C.sub.5-C.sub.30 carbocyclic group that is unsubstituted or substituted with at least one R.sub.10a, or two or more of the plurality of R.sub.20 are optionally linked to each other to form a C.sub.1-C.sub.30 heterocyclic group that is unsubstituted or substituted with at least one R.sub.10a, R.sub.10a is as defined for R.sub.1, * and *' each indicate a binding site to M.sub.1, a substituent of the substituted C.sub.5-C.sub.30 carbocyclic group, the substituted C.sub.1-C.sub.30 heterocyclic group, the substituted C.sub.1-C.sub.60 alkyl group, the substituted C.sub.2-C.sub.60 alkenyl group, the substituted C.sub.2-C.sub.60 alkynyl group, the substituted C.sub.1-C.sub.60 alkoxy group, the substituted C.sub.1-C.sub.60 alkylthio group, the substituted C.sub.3-C.sub.10 cycloalkyl group, the substituted C.sub.1-C.sub.10 heterocycloalkyl group, the substituted C.sub.3-C.sub.10 cycloalkenyl group, the substituted C.sub.1-C.sub.1 heterocycloalkenyl group, the substituted C.sub.6-C.sub.60 aryl group, the substituted C.sub.7-C.sub.60 alkyl aryl group, the substituted C.sub.7-C.sub.60 aryl alkyl group, the substituted C.sub.6-C.sub.60 aryloxy group, the substituted C.sub.6-C.sub.60 arylthio group, the substituted C.sub.1-C.sub.60 heteroaryl group, the substituted C.sub.2-C.sub.60 alkyl heteroaryl group, the substituted C.sub.2-C.sub.60 heteroaryl alkyl group, the substituted C.sub.1-C.sub.60 heteroaryloxy group, the substituted C.sub.1-C.sub.60 heteroarylthio group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group is each independently: deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, or a C.sub.1-C.sub.60 alkylthio group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, or a C.sub.1-C.sub.60 alkylthio group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.7-C.sub.60 alkyl aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.2-C.sub.60 alkyl heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —Ge(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.14)(Q.sub.15), —B(Q.sub.16)(Q.sub.17), —P(Q.sub.18)(Q.sub.19), —P(=O)(Q.sub.18)(Q.sub.19), or a combination thereof; a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.7-C.sub.60 alkyl aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.2-C.sub.60 alkyl heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group; a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.7-C.sub.60 alkyl aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.2-C.sub.60 alkyl heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —

CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.1-C.sub.60 alkylthio group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.7-C.sub.60 alkyl aryl group, a C.sub.7-C.sub.60 aryl alkyl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.2-C.sub.60 alkyl heteroaryl group, a C.sub.2-C.sub.60 heteroaryl alkyl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —Ge(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.24)(Q.sub.25), —B(Q.sub.26)(Q.sub.27), —P(Q.sub.28)(Q.sub.29), —P(=O)(Q.sub.28)(Q.sub.29), or a combination thereof; or —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —Ge(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.34)(Q.sub.35), —B(Q.sub.36)(Q.sub.37), —P(Q.sub.38)(Q.sub.39), or —P(=O)(Q.sub.38)(Q.sub.39), and Q.sub.1 to Q.sub.9, Q.sub.11 to Q.sub.19, Q.sub.21 to Q.sub.29, and Q.sub.31 to Q.sub.39 are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF.sub.5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.1-C.sub.60 alkylthio group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.7-C.sub.60 alkyl aryl group, a substituted or unsubstituted C.sub.7-C.sub.60 aryl alkyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1—Co heteroaryl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkyl heteroaryl group, a substituted or unsubstituted C.sub.2-C.sub.60 heteroaryl alkyl group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryloxy group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, or a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

2. The organometallic compound of claim 1, wherein M.sub.1 is Ir, and a sum of n₁ and n₂ is 3.
3. The organometallic compound of claim 1, wherein L.sub.1 is a ligand represented by one of Formulae 1A-1 to 1A-3: ##STR00111## wherein, in Formulae 1A-1 to 1A-3, X.sub.1 to X.sub.7, a₁, R.sub.20, and b₂₀ are each as described in claim 1, and * and *' each indicate a binding site to M.sub.1.
4. The organometallic compound of claim 1, wherein L.sub.1 is a ligand represented by Formula 1A(1): ##STR00112## wherein, in Formula 1A(1), R.sub.2 to R.sub.7, a₁, R.sub.20, and b₂₀ are each as described in claim 1, and * and *' each indicate a binding site to M.sub.1.
5. The organometallic compound of claim 1, wherein L.sub.1 is a ligand represented by one of Formulae 1A(1)-1 to 1A(1)-3: ##STR00113## wherein, in Formulae 1A(1)-1 to 1A(1)-3, X.sub.1, R.sub.2 to R.sub.7, a₁, R.sub.20, and b₂₀ are each as described in claim 1, and * and *' each indicate a binding site to M.sub.1.
6. The organometallic compound of claim 1, wherein L.sub.1 is a ligand represented by Formula 1A(2): ##STR00114## wherein, in Formula 1A(2), X.sub.1 to X.sub.7, R.sub.20, and b₂₀ are each as described in claim 1, and * and *' each indicate a binding site to M.sub.1.
7. The organometallic compound of claim 1, wherein L.sub.1 is a ligand represented by one of Formulae 1A(2)-1 to 1A(2)-6: ##STR00115## ##STR00116## wherein, in Formulae 1A(2)-1 to 1A(2)-6, X.sub.1, R.sub.2 to R.sub.7, a₁, R.sub.20, and b₂₀ are each as described in claim 1, and * and *' each indicate a binding site to M.sub.1.

8. The organometallic compound of claim 1, wherein R.sub.2 to R.sub.7, R.sub.20, and R.sub.31 to R.sub.33 are each independently: hydrogen, deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, or a C.sub.1-C.sub.20 alkylthio group; a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, or a C.sub.1-C.sub.20 alkylthio group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.10 alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or a combination thereof; a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzoimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group; a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzoimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a C.sub.1-C.sub.20 alkylthio group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl

group, a pyrimidinyl group, a pyridazinyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzoimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, or a combination thereof; or —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.4)(Q.sub.5), —B(Q.sub.6)(Q.sub.7), —P(Q.sub.8)(Q.sub.9), or —P(=O)(Q.sub.8)(Q.sub.9), and Q.sub.1 to Q.sub.9 are each independently: —CH.sub.3, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CH.sub.2CH.sub.3, —CH.sub.2CD.sub.3, —CH.sub.2CD.sub.2H, —CH.sub.2CDH.sub.2, —CHDCH.sub.3, —CHDCD.sub.2H, —CHDCDH.sub.2, —CHDCD.sub.3, —CD.sub.2CD.sub.3, —CD.sub.2CD.sub.2H, or —CD.sub.2CDH.sub.2; an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group; or an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group, each substituted with at least one of deuterium, a C.sub.1-C.sub.10 alkyl group, a phenyl group, or a combination thereof.

9. The organometallic compound of claim 1, wherein R.sub.2 to R.sub.7, R.sub.20, and R.sub.31 to R.sub.33 are each independently: hydrogen, deuterium, —F, —Cl, —Br, —I, —SF.sub.5, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, or a C.sub.1-C.sub.60 alkylthio group; or a group represented by any one of Formulae 9-1 to 9-39, 9-44 to 9-61, 9-201 to 9-237, 10-1 to 10-129, or 10-201 to 10-353; or —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3), or —N(Q.sub.4)(Q.sub.5), and Q.sub.1 to Q.sub.5 are each as described in claim 1: ##STR00117## ##STR00118## ##STR00119## ##STR00120## ##STR00121## ##STR00122## ##STR00123## ##STR00124## ##STR00125## ##STR00126## ##STR00127## ##STR00128## ##STR00129## ##STR00130## ##STR00131## ##STR00132## ##STR00133## ##STR00134## ##STR00135## ##STR00136## ##STR00137## ##STR00138## ##STR00139## ##STR00140## ##STR00141## ##STR00142## ##STR00143## ##STR00144## ##STR00145## ##STR00146## ##STR00147## ##STR00148## ##STR00149## ##STR00150## ##STR00151## ##STR00152## ##STR00153## ##STR00154## ##STR00155## ##STR00156## ##STR00157## ##STR00158## ##STR00159## ##STR00160## ##STR00161## wherein, in Formulae 9-1 to 9-39, 9-44 to 9-61, 9-201 to 9-240, 10-1 to 10-129, and 10-201 to 10-350, * indicates a binding site to a neighboring atom, “Ph” indicates a phenyl group, “TMS” indicates a trimethylsilyl group, and “TMG” indicates a trimethylgermyl group.

10. The organometallic compound of claim 1, wherein at least one of R.sub.2 to R.sub.7 is: —F, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3); a group represented by any one of Formulae 9-1 to 9-39 in which at least one hydrogen atom is substituted with —F, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3); or a group represented by any one of Formulae 10-1 to 10-129 or 10-201 to 10-350 in which at least one hydrogen atom is substituted with —F, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3): ##STR00162## ##STR00163## ##STR00164## ##STR00165## ##STR00166## ##STR00167## ##STR00168## ##STR00169## ##STR00170## ##STR00171## ##STR00172## ##STR00173## ##STR00174## ##STR00175## ##STR00176## ##STR00177## ##STR00178## ##STR00179## ##STR00180## ##STR00181## ##STR00182## ##STR00183## ##STR00184## ##STR00185## ##STR00186## ##STR00187## ##STR00188## ##STR00189## ##STR00190## ##STR00191## ##STR00192## ##STR00193## ##STR00194## ##STR00195## ##STR00196## ##STR00197## ##STR00198## ##STR00199## ##STR00200## ##STR00201## wherein, in Formulae 9-1 to 9-39, 10-1 to 10-129, and 10-201 to 10-350, * indicates a binding site to a neighboring atom, “Ph” indicates a phenyl group, “TMS” indicates a trimethylsilyl group, and “TMG” indicates a

trimethylgermyl group.

11. The organometallic compound of claim 1, wherein at least one of R.sub.2 to R.sub.7 is: —F, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3); a C.sub.1-C.sub.20 alkyl group that is substituted with at least one —F; or a group represented by one of Formulae 2A-1 to 2A-3: ##STR00202## wherein, in Formulae 2 Å-1 to 2 Å-3, R.sub.10 is hydrogen, deuterium, —F, —Cl, —Br, —I, —SF.sub.5, a hydroxyl group, a cyano group, a nitro group, an amino group, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a C.sub.1-C.sub.10 alkyl group, a C.sub.2-C.sub.10 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.1 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.1-C.sub.30 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —Ge(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.4)(Q.sub.5), —B(Q.sub.6)(Q.sub.7), —P(Q.sub.8)(Q.sub.9), or —P(=O)(Q.sub.8)(Q.sub.9), b₁₀ is an integer from 1 to 4, and Q.sub.1 to Q.sub.9 and Q.sub.31 to Q.sub.33 are each independently: —CH.sub.3, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CH.sub.2CH.sub.3, —CH.sub.2CD.sub.3, —CH.sub.2CD.sub.2H, —CH.sub.2CDH.sub.2, —CHDCH.sub.3, —CHDCD.sub.2H, —CHDCDH.sub.2, —CHDCD.sub.3, —CD.sub.2CD.sub.3, —CD.sub.2CD.sub.2H, or —CD.sub.2CDH.sub.2; an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group; or an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group, each substituted with at least one of deuterium, a C.sub.1-C.sub.10 alkyl group, a phenyl group, or a combination thereof.

12. The organometallic compound of claim 1, wherein R.sub.33 is hydrogen or deuterium.

13. The organometallic compound of claim 1, wherein the organometallic compound is represented by any one of Compounds 5-1 to 5-3: ##STR00203## wherein, in Formulae 5-1 to 5-3, M.sub.1, n₁, n₂, X.sub.1, R.sub.2 to R.sub.7, a₁, R.sub.20, b₂₀, and R.sub.31 to R.sub.33 are each as described in claim 1.

14. The organometallic compound of claim 1, wherein the organometallic compound is at least one of Compounds 1 to 72: ##STR00204## ##STR00205## ##STR00206## ##STR00207## ##STR00208## ##STR00209## ##STR00210## ##STR00211## ##STR00212## ##STR00213## ##STR00214## ##STR00215## ##STR00216## ##STR00217## ##STR00218## ##STR00219## ##STR00220## ##STR00221##

15. An organic light-emitting device, comprising: a first electrode; a second electrode; and an organic layer arranged between the first electrode and the second electrode, wherein the organic layer comprises an emission layer, and wherein the organic layer further comprises at least one of the organometallic compound of claim 1.

16. The organic light-emitting device of claim 15, wherein the emission layer comprises the at least one organometallic compound.

17. The organic light-emitting device of claim 15, wherein the emission layer further comprises a host, and an amount of the host in the emission layer is greater than an amount of the at least one organometallic compound in the emission layer, on a weight basis.

18. The organic light-emitting device of claim 16, wherein the emission layer emits a red light having a maximum emission wavelength of about 590 nanometers to about 650 nanometers.

19. The organic light-emitting device of claim 15, wherein the first electrode is an anode, the second electrode is a cathode, the organic layer further comprises a hole transport region arranged between the first electrode and the emission layer, and an electron transport region arranged between the emission layer and the second electrode, the hole transport region comprises a hole injection layer, a hole transport layer, an electron blocking layer, a buffer layer, or a combination thereof, and the electron transport region comprises a hole blocking layer, an electron transport layer, an electron injection layer,

or a combination thereof.

20. An electronic apparatus, comprising the organic light-emitting device of claim 15.
